

# **WEEK 4 – MACHINE LEARNING MODEL DEVELOPMENT**

Student Performance Prediction Using Regression Models

*Machine Learning Project Report*

## 1. Introduction

Machine Learning is a branch of Artificial Intelligence that enables computers to learn patterns from data and make predictions without being explicitly programmed for every task. In this project, regression techniques are applied to a student performance dataset to predict a student's Writing Score using demographic, educational, and academic features.

## 2. Project Objectives

The objectives of this project are to preprocess the student performance dataset, encode categorical variables, select appropriate features and a target variable, split the data into training and testing sets, develop Linear Regression and Decision Tree Regression models, evaluate their performance using standard regression metrics, compare the models, analyse errors and overfitting, interpret feature influence, and demonstrate a final prediction.

## 3. Dataset Description

The Students Performance dataset contains 1,000 student records. The variables used include Gender, Race/Ethnicity, Parental Level of Education, Lunch, Test Preparation Course, Math Score, Reading Score, and Writing Score. Writing Score was selected as the target variable.

## 4. Data Preprocessing

The dataset was inspected and categorical variables were converted into numerical representations through categorical encoding. The transformed data was then prepared for feature and target selection. This preprocessing step made the dataset suitable for regression algorithms.

## 5. Feature and Target Selection

Seven independent features were used: Gender, Race/Ethnicity, Parental Level of Education, Lunch, Test Preparation Course, Math Score, and Reading Score. The target variable was Writing Score. The feature matrix had shape (1000, 7) and the target vector had shape (1000,).

## 6. Train-Test Split

The dataset was divided into 800 training records and 200 testing records. The training feature matrix had shape (800, 7), while the testing feature matrix had shape (200, 7). This separation allowed the models to be evaluated on previously unseen data.

## 7. Linear Regression Model

Linear Regression was trained as the primary regression model. The model learned a linear relationship between the seven input features and Writing Score. The fitted intercept was 7.7492. The model coefficients were: Gender -5.5773, Test Preparation Course -3.2893, Reading Score 0.6894, Lunch 0.3818, Math Score 0.2655, Parental Level of Education -0.2103, and Race/Ethnicity 0.0688. Because categorical variables were numerically encoded, these coefficients should be interpreted cautiously and are not causal effects.

## 8. Decision Tree Regression Model

A Decision Tree Regression model was also trained using the same training and testing data. It was included as a non-linear model for comparison with Linear Regression.

## 9. Model Evaluation Metrics

The models were evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and  $R^2$  Score. Lower MAE, MSE, and RMSE values indicate smaller prediction errors, while an  $R^2$  value closer to 1 indicates stronger explanatory performance.

## 10. Model Comparison

Linear Regression achieved the lowest error values and the highest  $R^2$  score. Therefore, it was selected as the best-performing model for this project.

## 11. Actual vs Predicted Analysis

The Actual vs Predicted visualization showed that most predicted Writing Scores were distributed close to the diagonal reference line. This indicates that the Linear Regression model produced predictions that were generally close to the observed values.

## 12. Residual Analysis

Residual analysis was used to examine the difference between actual and predicted Writing Scores. The residuals were distributed around the zero reference line without a strong systematic pattern, supporting the model's overall fit.

## 13. Distribution of Prediction Errors

The prediction-error histogram showed that many errors were concentrated around the central region. This indicates that a large proportion of predictions were reasonably close to their actual values.

## 14. Overfitting / Underfitting Analysis

The Linear Regression model achieved a training  $R^2$  of 0.9418 and a testing  $R^2$  of 0.9312. The difference was only 0.0106. This small gap indicates that the model generalizes well to unseen data and that no significant overfitting was detected.

## 15. Feature Influence Analysis

The coefficient analysis showed the relative direction and magnitude of the model's learned relationships. Reading Score and Math Score had positive coefficients, while Gender and Test Preparation Course had negative coefficients in the encoded representation. These values describe the fitted model rather than causal relationships.

## 16. Final Prediction

A sample student record was passed to the selected Linear Regression model. The resulting predicted Writing Score was 75.75.

## 17. Results and Discussion

Both models produced good predictive performance, but Linear Regression was consistently better on the selected evaluation metrics. Its  $R^2$  score of 0.9312 indicates that approximately 93.12% of the variance in the test-set Writing Score was explained by the fitted model. The MAE of 3.3345 indicates an average absolute prediction error of about 3.33 score points.

## 18. Limitations

The project uses a limited set of available features and relies on numerical encoding for categorical variables. The model predicts Writing Score but does not establish causal relationships. Performance may change on a different student population or dataset.

## 19. Future Improvements

Future work could include one-hot encoding for categorical variables, cross-validation, hyperparameter tuning, feature selection, and comparison with Random Forest, Gradient Boosting, and other advanced regression algorithms. A larger and more diverse dataset could also improve robustness.

## 20. Conclusion

This project successfully completed the machine learning model development workflow for predicting student Writing Scores. Linear Regression was selected as the best-performing model, achieving MAE 3.3345, MSE 16.5880, RMSE 4.0728, and  $R^2$  0.9312. The training-testing  $R^2$  gap of 0.0106 indicates good generalization with no significant overfitting. The final sample prediction was 75.75.

### Model Evaluation Results

| Model                    | MAE    | MSE     | RMSE   | $R^2$ Score |
|--------------------------|--------|---------|--------|-------------|
| Linear Regression        | 3.3345 | 16.5880 | 4.0728 | 0.9312      |
| Decision Tree Regression | 3.9146 | 24.5673 | 4.9565 | 0.8981      |

Linear Regression is the best-performing model because it has the lowest MAE, MSE, and RMSE, and the highest  $R^2$  score.

### Linear Regression Coefficients

| Feature                 | Coefficient |
|-------------------------|-------------|
| Gender                  | -5.5773     |
| Test Preparation Course | -3.2893     |
| Reading Score           | 0.6894      |
| Lunch                   | 0.3818      |
| Math Score              | 0.2655      |

|                             |         |
|-----------------------------|---------|
| Parental Level of Education | -0.2103 |
| Race/Ethnicity              | 0.0688  |

## Final Project Summary

| Component               | Result                       |
|-------------------------|------------------------------|
| Dataset                 | Students Performance Dataset |
| Total Records           | 1,000                        |
| Features                | 7                            |
| Target                  | Writing Score                |
| Training / Testing      | 800 / 200                    |
| Best Model              | Linear Regression            |
| Best $R^2$ Score        | 0.9312                       |
| Best MAE                | 3.3345                       |
| Best RMSE               | 4.0728                       |
| Training $R^2$          | 0.9418                       |
| Testing $R^2$           | 0.9312                       |
| $R^2$ Difference        | 0.0106                       |
| Final Sample Prediction | 75.75                        |

Recommended figures for the final submission: Actual vs Predicted plot, Residual Plot, Prediction Error Distribution, Feature Influence plot, and Model Comparison charts (MAE, RMSE, and  $R^2$ ). These can be inserted from the Colab notebook if required.